"Optimizing E-Commerce Customer Spending: A Predictive Analytics Approach for TokoBli Campaign Strategy"

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1. Business Understanding

Problem Statement:

Which factors have a significant influence on the amount customers spend; these factors will subsequently be used to develop business campaigns and strategies aimed at increasing customer spending.

The Metrics that use for answer Problem Statement above

Link to Dataset

Customer Id	Gender	Age	Income	Tenure	Avg. Session Time	Total Promo	Bounce Rate	Yearly Spending
100	M	30	12,000,000	10	520	12	56	4,000,000
101	F	30	8,600,000	7	744	9.46	57	4,300,000
102	M	30	7,200,000	3	114	22	40	4,800,000
103	F	30	30,000,000	9	169	18	37	5,500,000
104	M	20	10,440,000	14	736	13.5	36.2	5,800,000
105	F	20	18,900,000	9	1104	8.8	30.7	6,300,000
106	M	20	13,200,000	5	161	26.4	31	6,600,000
107	F	21	10,650,000	10	212	25.56	21.9	7,100,000
108	M	33	10,140,000	18	976	21	15	7,800,000
109	F	34	14,580,000	11	1428	16.5	10.9	8,100,000
110	M	35	13,200,000	7	213	34.4	5	8,600,000
111	F	36	10,650,000	12	271	33.48	25	9,300,000
112	M	20	18,900,000	21	1192	28.8	25	9,600,000
113	F	21	14,580,000	13	1788	22.22	18.9	10,100,000
114	M	22	13,200,000	9	270	43.2	11.2	10,800,000
115	F	33	10,650,000	7	128	14.4	56	4,000,000
116	M	30	8,600,000	11	278	12.9	52.7	4,300,000
117	F	30	30,000,000	6	170	5.5	67	3,000,000
118	M	30	10,440,000	1	225	12	68	3,300,000
119	F	30	18,900,000	7	246	13.68	58.2	3,800,000

The Factors will be check with correlation are : variables **Age, Income, Tenure, Avg. Session Time, Total Promo, and Bounce Rate**, which variables have a strong relationship with each other

2. Correlation

which variables have a strong relationship with each other?

Pearson's Method used cause type of Variabel data to be measure are interval and ratio

Pearson Test (two-sided) Correlation matrix

	Age	Income	Tenure	Avg. Session Tir	Total Promo	Bounce Rate
Age		0.141*	0.160**	0.193***	0.208***	-0.129*
Income	0.141*		0.379***	0.484***	0.446***	-0.448***
Tenure	0.160**	0.379***		0.685***	0.621***	-0.570***
Avg. Session Time	0.193***	0.484***	0.685***		0.777***	-0.787***
Total Promo	0.208***	0.446***	0.621***	0.777***		-0.794***
Bounce Rate	-0.129*	-0.448***	-0.570***	-0.787***	-0.794***	

Significance: *<0.05,**<0.01,***<0.001

Pearson Test Correlation data

Pair	df	r	p	CI 0.95 Lower	CI 0.95 Upper
Income|Age	297	0.14077443	0.01484391	0.02778783	0.25020939
Tenure|Age	297	0.16034759	0.00545166	0.04778642	0.26888731
Tenure|Income	297	0.37894671	0	0.27744197	0.47208491
Avg. Session Time|Age	297	0.19344527	0.00077159	0.08181007	0.30028654
Avg. Session Time|Income	297	0.48369206	0	0.39175554	0.56606503
Avg. Session Time|Tenure	297	0.68533196	0	0.62010717	0.74114739
Total Promo|Age	297	0.20780328	0.00029715	0.09665114	0.31383611
Total Promo|Income	297	0.44567271	0	0.34993246	0.53219889
Total Promo|Tenure	297	0.62137447	0	0.54646014	0.68642379
Total Promo|Avg. Session Tim	297	0.77688616	0	0.72757119	0.8182135
Bounce Rate|Age	297	-0.12897538	0.0257359	-0.23891051	-0.01577586
Bounce Rate|Income	297	-0.4483756	0	-0.53461569	-0.35289321
Bounce Rate|Tenure	297	-0.57027661	0	-0.64216727	-0.48844197
Bounce Rate|Avg. Session Tir	297	-0.78730561	0	-0.82689099	-0.73995662
Bounce Rate|Total Promo	297	-0.79414346	0	-0.83257543	-0.74810214

	Very Strong Correlation
	Strong Correlation
	Moderate Correlation
	Weak Correlation

Strong Relationship:

1. Avg.Session Time Vs Tenure (Positive Correlation)
2. Total Promo Vs Tenure (Positive Correlation)
3. Bounce Rate Vs Tenure (Negative Correlation)

Very Strong Relationship

1. Total Promo Vs Avg. Session Time (Positive Correlation)
2. Bounce Rate Vs Avg. Session Time (Negative Correlation)
3. Bounce Rate Vs Total Promo (Negative Correlation)

Insight:

1. Total Promo, Avg. Session Time, and Bounce Rate are tightly linked. Customers who use more promos spend longer on the site and leave less often.
2. Tenure builds engagement. Loyal (long-tenure) customers browse longer, use more promos, and bounce less. Retention matters.
3. Bounce Rate is the "mirror" variable. Every other variable correlates negatively with it, so it works as an indicator of poor engagement.

Note: The correlation matrix shown above displays the relationships between variables/factors (e.g., Promo vs. Bounce Rate). It does not show the relationship between each individual variable/factor and Spending. Therefore, the next step is to check for multicollinearity among all these variables/factors.

3. Regresion

What variable has a significant impact on the Yearly Spending?

Multiple Linear Regression to predict Yearly Spending							
Model Summary							
Predictors	Age,Income,Tenure,Avg. Session Time,Total Promo,Bounce Rate						
R	0.96399072						
R Square	0.92927811						
Adjusted R Squa	0.92782492						
Std. Error of the	1384406.116						
Sample size	299						
Calculated with lambda 0							
Step 1							
ANOVA							
Source	SS	df	MS	F	Sig.		
Regression	7.35363E+15	6	1.2256E+15	639.4747754	0		
Residual	5596414460075	292	1916580294546				
Total	7.91327E+15	298					
Coefficients							
Variable	Intercept	Age	Income	Tenure	Avg. Session Tir	Total Promo	Bounce Rate
B	1947879.706	30555.49209	0.03582464	64158.04723	6286.982772	133215.6971	-50349.9385
StdError	594065.0494	16729.0723	0.0111173	14092.06659	456.2684441	13692.77277	6451.813391
Beta		0.02926946	0.05808334	0.09936873	0.41779952	0.27971716	-0.22496696
t	3.27889969	1.82649053	3.2224224	4.55277775	13.77913124	9.72890585	-7.80399795
p	0.0011682	0.06879731	0.00141493	0.00000778	0	0	0
VIF		1.06028719	1.34142762	1.9668696	3.79595587	3.41302277	3.43109031
Karena Nilai P.Value Variabel Age >	Maka lanjut ke iterasi 1						

Variabel that has a significant impact for yearly Spending are :

1. Income
2. Tenure
3. Avg. Session Time
4. Total promo
5. Bounce Rate

Uji Simultan	
HO	semua varibel independen tidak memiliki dampak terhadap variabel dependent
H1	minimal ada satu varibel independen yang memiliki dampak terhadap variabel dependent
Alpha	5%
P value	0
P.Value< Alpha	
Conclusion	H1
Uji Parsial	
Ho	Var.dependent tidak memiliki dampak terhadap var.Independent
H1	salah satu Var.dependent memiliki dampak terhadap var.Independent

3. Regresion

Iterasi 1 ⟹ Mathematic Formula For Prediction?

Multiple Linear Regression to predict Yearly Spending

Model Summary						
Predictors	Income,Tenure,Avg. Session Time,Total Promo,Bounce Rate					
R	0.96357154					
R Square	0.92847012					
Adjusted R Squa	0.92724947					
Std. Error of the	1389914.044					
Sample size	299					
Calculated with lambda 0						
Step 1						
ANOVA						
Source	SS	df	MS	F	Sig.	
Regression	7.34723E+15	5	1.46945E+15	760.6380056	0	
Residual	5660352875207	293	1931861049559			
Total	7.91327E+15	298				
Coefficients						
Variable	Intercept	Income	Tenure	Avg. Session Tin	Total Promo	Bounce Rate
B	2590241.49	0.03684317	64576.21329	6341.901705	136147.7235	-49209.12635
StdError	480680.5048	0.01114748	14146.26528	457.0879684	13652.45109	6447.057484
Beta		0.05973472	0.10001639	0.42144914	0.28587363	-0.21986973
t	5.38869678	3.305067	4.56489483	13.87457589	9.97240148	-7.63280403
p	0.00000015	0.00106758	0.00000736	0	0	0
VIF		1.33805254	1.96635045	3.77947085	3.36611365	3.39893431

indicates that a customer's Yearly spendingis almost entirely (92.7%) influenced by the 5 independent variables in the model (Income, Tenure, Avg. Session Time, Total Promo, and Bounce Rate)

Uji Simultan	
HO	semua varibel independen tidak memiliki dampak terhadap variabel de
H1	minimal ada satu varibel independen yang memiliki dampak terhadap variabel dependent
Alpha	5%
P value	0
P.Value< Alpha	
Conclusion	H1

Uji Parsial	
Ho	Var.dependent tidak memiliki dampak terhadap var.Independent
H1	salah satu Var.dependent memiliki dampak terhadap var.Independent

The p-values for all dependent variables are smaller than the alpha value (5%), and the VIF values are below 5.

Yearly Spending = 2590241.49 + (0.036 x Income) + (64576.21 x Tenure) + (6341.90 x Avg. Session Time) + (136147.72 x Total Promo) - (49209.12 x Bounce Rate)

3. Regresion

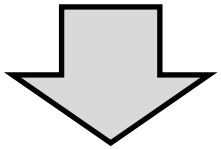
Prediction Yearly Spending for new Costumer with data :
Age (20), Income (20,000,000), Tenure (1), Avg. Session Time (1000), Total Promo (20), Bounce Rate (30)

Formula:
Yearly Spending = 2590241.49 + (0.036 x Income) + (64576.21 x Tenure) + (6341.90 x Avg. Session Time) + (136147.72 x Total Promo) - (49209.12 x Bounce Rate)

INSIGHT

1. Every 1 rupiah increase in a customer's income, Yearly Spending rises by Rp 0,036.
2. Each 1 month increase in membership duration (tenure) will raise Yearly Spending by Rp 64.576,21.
3. Every 1 second increase in Avg. Session Time, Yearly Spending rises by Rp 6.341,90.
4. Each 1 % increase in Total Promo will raise Yearly Spending by Rp 136.147,72.
5. For every 1% increase in the bounce rate, Yearly Spending drops by Rp 49.09,12.

Potential Yearly Spending:
Yearly Spending = 2590241.49 + (0.036 x 20.000.000) + (64576.21 x 1) + (6341.90 x 1000) + (136147.72 x 20) - (49209.12 x 30)



Rp 10.963.398,50

4. Recomendation

Objective	Business Impact	Isolation (Root Cause)	Prioritization	Recommendation
Reduce Bounce Rate	Every bounce immediately cuts down yearly spending significantly.	Users leave because the landing page fails to grab attention.	P1 (Urgent)	• Place main promo banners at the very top of the page (*Above the Fold*). • Deploy an automatic discount *Exit-Intent Pop-up* when a user tries to leave.
Maximize Revenue per Promo	Promotions are the single strongest driver of yearly spending.	Current promos are flat discounts that do not incentivize higher spending per order.	P2 (High)	Shift from flat discounts to minimum purchase thresholds (e.g., Rp 20.000 off on orders over Rp 100.000) to boost Average Order Value.
Retain Long-Term Customers	Established users naturally generate stable, high lifelong spending.	Risk of loyal users getting bored and switching to competitors if left unappreciated.	P3 (Medium)	• Launch a tiered loyalty program (Silver, Gold, Platinum). • Automatically send a voucher on the account creation anniversary (Anniversary Reward).
Increase Session Duration	The longer users browse, the higher the likelihood of a checkout.	he platform is currently too transactional and lacks engaging features to keep users browsing.	P4 (Low)	• Introduce simple gamification (e.g., Daily Check-ins for shopping points). • Optimize the Cross-Selling product recommendation engine at the bottom of pages